[SIEM Fundamentals for SOC L1 Analyst]

[1. What is SIEM]

SIEM stands for **Security Information and Event Management**.

> *"SIEM helps security teams monitor, detect, and investigate cyber threats centrally using unified system logs."*

A SIEM is a security solution that:

- Collects logs from completely different enterprise devices.
- Stores logs in one centralized data repository.
- Detects suspicious activity by cross-examining logs.
- Generates real-time security alerts.
- Helps SOC analysts quickly investigate and scope incidents.

[Why SIEM is Important]

SIEM is vital because it allows teams to detect attacks quickly, monitor all enterprise systems centrally, perform end-to-end incident investigations, create automated alerts, maintain rigorous compliance audits, and improve high-level visibility across networks.

[Connected Infrastructure Ecosystem]

```
[ Log Sources Ecosystem ]
   ├── Perimeter & Network ──> Firewalls, Routers, Switches, IDS/IPS
   ├── Host & Server Line ──> Windows Systems, Linux Servers, Web Servers
   └── Security & Cloud   ──> Antivirus/EDR Engines, Managed Cloud Services
```

[Core SIEM Pipeline Workflow]

```
Log Sources ──> SIEM Ingestion ──> Correlation Engine ──> Triggered Alerts ──> Analyst Investiga
```

[2. Log Collection]

Definition: The systematic process of collecting raw event logs from diverse architectural devices and software applications directly into the central SIEM parser engine.

[Common Log Ingestion Mapping]

Log Source	Practical Log Example Details
Firewall	Blocked external connections, traffic drops, port violation metrics.
Windows	User logon tracking, authentication events, account updates.
Linux	SSH connection logs, sudo command usage tracking, daemon states.
Antivirus / EDR	Malware detection triggers, file signature blocks, quarantined elements.
Proxy Server	Internal website access tracking, URL request categories.
Web Server	HTTP response status codes, raw request lines, web method tracking.

[Log Collection Methods]

```
[ Collection Methods ]
   ├── Agent-Based Collection ──> Software installed on endpoint host (e.g., Splunk Forwarder, W
   └── Agentless Collection   ──> Logs pushed/pulled remotely via standard hooks (e.g., Syslog,
```

[Common Log Protocols]

- **Syslog:** Standard network mechanism used for Linux systems and routing appliances.
- **SNMP:** Used for tracking performance metrics and monitoring structural system health.
- **WMI (Windows Management Instrumentation):** Native method for collecting deep Windows host logs.
- **API Connections:** Used to stream cloud-native audit logs down into the monitoring matrix.

[3. Parsing & Normalization]

[Parsing]

The technical procedure of converting a chaotic text string or unstructured raw log file into clearly mapped, indexed, and readable metadata fields.

Example Raw Log String:

```
User=admin IP=192.168.1.5 Status=Failed
```

Parsed Field Matrix:

Extracted Key Field	Assigned Indexed Value
`User`	admin
`IP`	192.168.1.5
`Status`	Failed

[Normalization]

The practice of taking mismatched field names produced by different infrastructure vendors and mapping them uniformly into a single common format standard identifier.

```
[ Mapped Vendor Strings ]
   ├── Vendor A Log Key: "src_ip"    ──┐
```

```
  ├── Vendor B Log Key: "sourceip"  ───┼───> [ Normalized Schema Target Field ] ──> Source_IP
  └── Vendor C Log Key: "client_ip" ───┘
```

Why Parsing & Normalization are Crucial: They enable fast querying, permit seamless rule correlation execution across multiple vendor sets, reduce false positive alert rates, and accelerate active analyst security investigations.

[4. Correlation Rules]

Definition: The logic module inside a SIEM that connects separate events occurring over time across different infrastructure nodes to pinpoint hidden attack patterns.

Practical Scenario Mapping:

```
[ Ingestion Timeline ]
  Event 1: Failed Login (Host A)
  Event 2: Failed Login (Host A)
  Event 3: Failed Login (Host A)  ───> [ SIEM Correlation Rule Engine ] ───> Alert Triggered:
  Event 4: Failed Login (Host A)           (Threshold Match: 5 Attempts)        [ Possible Brut
  Event 5: Failed Login (Host A)
  Event 6: SUCCESSFUL Login (Host A)
```

[Common Correlation Rules Baseline]

Rule Condition Trigger Matrix	Correlated Defensive Threat Deduction
High number of failed logons followed immediately by a single success.	Possible Brute Force / Account Compromise
Single user account logging in from two countries within short intervals.	Impossible Travel / Session Theft
Local endpoint triggers a malware signature alert followed by immediate unknown outbound connections.	Host Compromise / C2 Callbacks
Administrative service account initiating login requests at midnight hours.	Suspicious Privilege Abuse / Rogue Actor Activity

Core Elements of a Rule: Conditions (logic statements), Thresholds (numeric bounds), Time Window (temporal scope), Severity (impact rating), and Action (automated notification execution).

[5. Dashboards]

Definition: The unified visual telemetry console showing high-level overviews of live security events, baseline deviations, and pending analysis logs.

Operational Use Cases: Real-time security state tracking, direct triage queue visibility, threat trend analysis, and automated leadership metric report compilation.

[Common UI Telemetry Elements]

- **Pie Charts:** Grouping alert categories by percentage split.
- **Trend Graphs:** Visualizing log generation volume trends over time windows.
- **Recent Alert Tables:** Displaying critical triage items waiting in the queue.
- **Geo-IP Mapping Modules:** Highlighting physical source locations of boundary block alerts.

Standard L1 Monitoring View: Tracks top attacking source IPs, real-time failed logons, critical malware signatures, severe unassigned alert volumes, and overall user activity trends.

[6. Alerting]

Definition: The automatic generation of priority notification records pushed to the analyst triage queue when correlation criteria are met.

[Triage Severity Matrix]

Severity Tier	Operational Security Meaning & Response Posture
Low	Minor infrastructure issue, behavioral divergence, low-risk event logging.
Medium	Suspicious system event matching anomaly baselines, requires monitoring review.
High	Serious validated threat profile matching attacker tactics, requires prompt escalation.
Critical	Active system compromise, live ransomware deployment, or confirmed breach in progress.

Common Alarm Classifications: Brute force detections, unexpected local malware triggers, unauthorized privilege escalations, suspicious network exfiltration volumes, and boundary access control violations.

Alert Lifespans Workflow: Event Extraction ——> Rule Matching ——> Alert Entry Generation ——> L1 Analyst Queue Ingestion & Live Investigation.

[7. Search Queries]

Definition: Specialized conditional command strings executed by analysts to query, parse, and isolate logs within the SIEM architecture during investigations or proactive operations.

[Essential Logic Filter Syntaxes]

```
1. Failed Authentications Lookup:   status="failed"
2. Isolated System IP Check:         src_ip="192.168.1.10"
3. Targets User Tracking:           user="admin"
4. Specific Event Category Hunt:     event_type="malware"
```

Analysts use these queries to drill down into anomalous timeline items, strip away background system noise, isolate malicious entities, and hunt for hidden indicators.

[8. Splunk Basics]

Definition: A leading commercial enterprise data processing platform used extensively for SIEM log indexing, parsing, and analysis.

[Structural Architecture Model]

```
Log Source Engine ——> Forwarder (Ships Data) ——> Indexer (Stores/Parses) ——> Search Head (User
```

[Fundamental Splunk SPL Query Examples]

Dump All Stream Content:

```
index=*
```

Audit Failed Local Logons (Event ID 4625):

```
index=windows EventCode=4625
```

Isolate Target Attacking Node IP:

```
src_ip=192.168.1.5
```

[Important SPL Commands Reference]

SPL Command	Functional Use Case Description
`search`	The foundational command used to filter raw events matching string conditions.
`stats`	Calculates statistical values (counts, averages) across targeted search variables.
`table`	Reformats specified metadata fields into a clean visual spreadsheet table.
`sort`	Orders returned log queries chronologically or numerically.
`timechart`	Generates structured timeline data plots optimized for dashboard visualization.

Practical Analytical Query Sample:

```
index=windows EventCode=4625 | stats count by user
```

Analytical Meaning: Evaluates total failed Windows logon events across the network and organizes them into an ordered breakdown by username to catch targeted brute-force activity.

[9. Elastic Stack Basics]

Also known as the **ELK Stack**: an open-source log aggregation ecosystem consisting of Elasticsearch, Logstash, and Kibana.

[Architectural Workflow Model]

```
Raw Log Data ──> Logstash (Ingests & Parses) ──> Elasticsearch (Indexes & Stores) ──> Kibana (V
```

[Modular Beats Ingestion Architecture]

- **Filebeat:** Lightweight local utility designed explicitly to ship flat server text logs.
- **Winlogbeat:** Tailored specifically to extract internal Windows event logs into the pipeline.
- **Packetbeat:** Intercepts real-time network protocol layer data to monitor traffic behaviors.

Kibana Workspace Purpose: Provides tier-1 SOC operators with visualization dashboards, interactive query management screens, configuration control points, and threat hunting views.